

CIL: Does This Anomaly Score Belong to This Video?

Content-Identity Auditing of Temporal Score Curves

Author Names Required Before Submission

Abstract—Temporal anomaly scores can agree with labels even when the curve paired with a video is unnecessary: transferable progress patterns or video-specific side information may yield similar ranking utility. We introduce Content Identity Licensing (CIL), a fail-closed offline ownership audit for completed temporal score curves. CIL measures target-identity advantage (TIA), the video-macro average-precision gain of a target’s own rank curve over progress-matched donor curves, and subtracts the strongest predeclared content-free sentinel to obtain a content-necessity margin (CNM). Here content identity denotes only own-curve advantage beyond named sentinels under the donor rule, not pixel or semantic attribution. Frozen effect-size, bootstrap, strata, label-flow, and replay gates must all pass. Three prospectively sealed settings support this bounded construct. OOPS latent motion attains CNM 0.0575 on 256 untouched real-camera videos; a train-only Pistachio ridge score attains CNM 0.1113 on 120 mixed-label targets from 257 held-out synthetic videos. A one-shot, label-firewalled IITB Corridor evaluation reports a complete three-candidate real-surveillance family: all candidates pass simultaneous CIL gates on 134 targets and ordinary-utility gates on all 150 test videos, with temporal novelty attaining CNM 0.0651, AUC 0.7164, and AP 0.7523. Wrong-owner, side-information, failed-candidate, and low-utility controls show why a license cannot be interpreted as semantic understanding, visual causality, or detector superiority. A post-confirmation two-way target/donor bootstrap retains positive CNM for OOPS, Pistachio ridge, and every IITB candidate.

Index Terms—Video anomaly detection, multimedia evaluation, temporal score curves, counterfactual auditing, reproducibility.

I. INTRODUCTION

FRAME- and video-level ranking metrics answer whether anomaly scores agree with labels, but not whether the completed score curve paired with a target video matters. A detector can receive high utility from a common progress pattern, duration, sampling schedule, or another video-specific side channel. The curve may then be useful but largely transferable between videos. We ask a narrower multimedia-evaluation question: *does this completed temporal score belong to this video?*

Utility and ownership are distinct: a shared progress curve can rank labels well yet has zero TIA because every donor reproduces it. This instance-level gap parallels broader concerns about shortcut learning and underspecified machine-learning pipelines [25], [26].

We introduce Content Identity Licensing (CIL), an ownership counterfactual that holds target labels fixed while replacing the target’s score ranks with progress-normalized donor

ranks. Target-identity advantage (TIA) alone is insufficient because nonvisual side information can also produce an identity advantage. CIL therefore subtracts the strongest member of a frozen sentinel family and grants a license only when the content-necessity margin (CNM), effect-size, uncertainty, strata, label-flow, complete-reporting, and replay gates pass.

CIL is an offline audit of an already completed curve, not an online calibration rule or a new detector. Within-video ranking deliberately removes score scale to isolate temporal ownership; ordinary AP and AUC remain necessary for utility. Our evidence is consequently falsification-oriented: three sealed confirmations, deterministic wrong-owner reassignment, crossed target/donor uncertainty, a nonvisual counterexample, failed frozen candidates, modern-model pressure tests, and a complete retrospective surveillance family.

The contributions are:

- a model-agnostic ownership estimand for completed temporal score curves, distinct from both detector construction and ordinary ranking utility;
- a conservative sentinel-relative margin and fail-closed protocol that freezes candidates, donors, thresholds, data roles, and claim boundaries;
- three prospective confirmations, including a one-shot multi-sentinel real-surveillance family with simultaneous inference and ordinary-utility promotion gates; and
- explicit negative, robustness, reproducibility, data-rights, and deployment boundaries that prevent a bounded license from becoming a semantic or causal claim.

II. RELATED WORK

Video anomaly detection (VAD) conventionally reports ranking utility [1]–[3]. Complementary evaluation work shows that AUROC may hide low-false-positive recall [4]; range, event, region, and attribute metrics expose temporal or spatial failures hidden by frame scores [5]–[8]; and semantic benchmarks measure grounding beyond scalar localization [9]. Multi-domain evaluation further reveals incompatible definitions of abnormality [10]. CIL replaces none of these metrics. It asks whether a finished score curve carries target-video-specific ranking information.

Benchmark audits advocate reliable measures, simple baselines, and consistent protocols [11], [12]. Ji *et al.* remove video-wise min–max normalization in favor of train-normal score calibration to avoid an unrealistic per-video test-time operation [13]. Three recent VAD audits sharpen adjacent

failure modes. Song and Lee decompose pooled frame metrics into within- and cross-video comparisons [27]; Abdulaziz and Bondarev audit global, category, and within-video AUC granularity and scene bias [28]; and Rashidi measures cross-dataset surveillance transfer [29].

These studies audit metric composition, score normalization, or transfer across scenes. CIL tests a different counterfactual: after fixing the target labels, it replaces only the owner of an already completed rank curve with progress-matched foreign owners and then subtracts a frozen sentinel maximum. It therefore does not estimate pooled-metric dilution, normalization-induced calibration error, or cross-dataset deployability. Conversely, those audits do not test whether the target's own curve is necessary relative to foreign curves on the same labels under a prospective information barrier. The questions are complementary rather than competing leaderboard metrics.

Permutation-based VAD methods reorder frames or discriminate adjacent windows to *construct* scores [14]–[16]. CIL instead reassigns finished curve ownership and evaluates own and foreign curves on the same target labels. It uses permutation-test logic [17] without claiming to invent permutation testing. Because frames are correlated and donors are reused, our independent unit is video and we use video-level and crossed target/donor resampling [18], [19].

III. CONTENT IDENTITY LICENSING (CIL)

A. Ownership Substitution

Let video i have temporal labels $y_i = (y_{it})_{t=1}^{T_i}$ and a completed score curve $s_i = (s_{it})_{t=1}^{T_i}$. Each curve is transformed to tie-aware within-video percentile ranks $r_i(u)$ at normalized progress $u \in [0, 1]$. This discards absolute score scale and permits transfer between videos of different lengths. Let $D(i)$ be a deterministic, predeclared donor set and A a per-video ranking functional; we use temporal average precision. Then

$$\begin{aligned} \text{TIA}_i(s) &= A(r_i, y_i) - \frac{1}{|D(i)|} \sum_{j \in D(i)} A(r_j \rightarrow i, y_i), \\ \text{TIA}(s) &= \frac{1}{n} \sum_i \text{TIA}_i(s), \end{aligned} \quad (1)$$

where $r_j \rightarrow i$ is donor j interpolated to target progress. Positive TIA means that own-curve ranks outperform the frozen donor curves; it does not distinguish visual evidence from ownership-specific proxies.

B. Sentinel-Relative Margin

Before scientific labels are opened, we specify content-free sentinel curves $\mathcal{Q}$, such as center-time, duration/progress, linear-progress, and sampling-grid priors. The conservative multi-sentinel margin is

$$\text{CNM}(s) = \text{TIA}(s) - \max_{q \in \mathcal{Q}} \text{TIA}(q). \quad (2)$$

The maximum is recomputed within every bootstrap draw. CNM is conservative only relative to the named family; unmeasured shortcuts remain outside its identification scope.

C. Decision Rule and Statistical Unit

CIL grants a license only if marginal video-bootstrap lower bounds of TIA and CNM exceed zero, frozen point-effect floors and fixed strata conditions pass, and prediction, label-flow, replay, and complete-reporting gates pass. When a candidate family is tested jointly, max-deviation one-sided lower bounds control the complete family under the bootstrap approximation. The resampling unit is video rather than frame. Crossed target/donor resampling provides a secondary sensitivity analysis; its addition after confirmation does not change the original decision rule.

The donor shifts are offsets in a deterministic hash ordering of video IDs, not offsets in time. The common list $\{7, 19, 31, 43, 59, 73, 91, 109\}$ supplies eight distinct non-self donors spread through that ordering; modulo reduction is collision-checked for each role. Neither the order nor the shifts use scores or labels. Eight was fixed as a finite compromise between donor coverage and repeated-evaluation cost, not as an estimated optimum. Point-effect floors are likewise protocol thresholds in video-macro AP units, not universal constants: they require a nontrivial effect in addition to a positive interval. Their exact study-specific values and the fact that they were locked before the corresponding confirmation are reported in the supplement.

Average within-video ranks make the audit invariant to strictly increasing recalibration. TIA is exactly zero when each transferred donor rank vector equals the target rank vector on the target grid. The expected wrong-owner contrast is zero under conditional ownership exchangeability. These are properties of the fixed construction, not proof that donors are semantically exchangeable.

D. Finite-Sample Properties and Identification

The audit has five properties that make its output interpretable while delimiting its claim.

Monotone invariance: temporal AP depends only on ordering. Applying any strictly increasing transform separately to every candidate curve leaves each own and donor AP term, and therefore TIA and CNM, unchanged. This removes score scale from the ownership question, but prevents CIL from measuring calibration.

Shared-curve null: if each transferred donor rank vector equals the own rank vector on the target grid, the own and donor AP terms coincide and TIA is exactly zero. Equal-length copies of one ranked template satisfy this condition. A common continuous template sampled at different lengths need not: finite-grid ranking and interpolation can change ties or ordering. Such discretization effects are measured by the named sentinels rather than assumed absent.

Wrong-owner exchangeability: under exchangeability of score ownership conditional on the normalized progress grid, the expected contrast between the own curve and a randomly reassigned owner is zero. Our cyclic wrong-owner map is deterministic and reproducible; it is a finite-sample falsifier rather than a proof that exchangeability holds.

Named-family conservatism: because CNM subtracts the maximum sentinel TIA within every draw, it is no larger than

the margin against any individual named sentinel. This protects against selecting a convenient weak control. It does not protect against an omitted sentinel or an unrecorded data-generation shortcut.

Crossed dependence: a finite donor pool is reused across target videos, which couples target effects. In the crossed analysis, we retain the eight observed donor edges per target, draw independent multinomial target and unique-donor weights, and recompute the weighted mean on those edges. Unused target–donor pairs are not filled in. This two-way pigeonhole bootstrap [19] probes sensitivity to variation in both roles; it does not establish exact coverage for this graph. The prospective decision remains bound to each study’s original frozen protocol; crossed resampling is labeled as secondary when added after confirmation.

Identification consequently relies on videos being the independent sampling units, the official label grid defining the observation scale, progress normalization making transfer meaningful enough for the audit, and target and donor videos being sufficiently exchangeable under the frozen role partition. Positive TIA identifies own-versus-foreign temporal alignment under those conditions. Positive CNM additionally exceeds the named controls. Neither quantity identifies anomaly semantics or the causal source of the curve.

E. Operational Fail-Closed Protocol

Figure 1 summarizes the CIL estimand and its operational state machine. Candidate construction, score orientation, sentinels, donors, bootstrap seed and draw count, effect floors, strata, sample-size floors, utility gates, and permitted wording are frozen before confirmation. Label-free predictions are exported twice into deterministic archives. The confirmation role opens only if the archives are byte-identical and record zero label access. Evaluation then reports the complete family; a failed candidate cannot be removed and a passing curve cannot be inverted, smoothed, fused, or retuned after label access. Finally, manuscript values and figures must trace to the sealed result rather than an exploratory notebook.

This design separates three evidence tiers. A *prospective confirmation* requires a genuine information barrier and one-shot role opening. A *post-confirmation robustness analysis* may probe a consumed endpoint but cannot become another independent confirmation. A *retrospective stress test* may pre-freeze analysis rules on existing arrays but cannot recreate unread labels. We retain all tiers because collapsing them would reward post-outcome adaptation.

IV. PROSPECTIVE STUDIES AND EVIDENCE ROLES

A. OOPS and Pistachio

The historical OOPS study [20] froze one V-JEPA2 latent-motion candidate and one middle-time sentinel. It used 128 source videos to qualify the endpoint and 256 disjoint, unopened confirmation videos. Primary and replay outputs were byte-identical. OOPS is an in-the-wild real-camera transition-localization setting, and its original dataset exposes a middle-time prior; we preserve its single-sentinel contract rather than retroactively rewrite it.

TABLE I
FROZEN STUDY ROLES AND THEIR PERMITTED EVIDENCE USE.

Study	Source	Test	Sentinels	Evidence role
OOPS	128	256	middle time	prospective, real camera
Pistachio	65	120	four-sentinel max	prospective, synthetic
IITB Corridor	208 train	150 test	four-sentinel max	prospective, surveillance
ShanghaiTech	–	53	four-sentinel max	retrospective stress test

Pistachio [21] is a synthetic, balanced, long-form VAD/VAU benchmark. Its fixed train-only linear ViT family uses 65 evaluable source and 120 mixed-label targets within a 257-video held-out confirmation role. The remaining 137 constant-label videos are excluded from the ownership target mean but remain eligible donors; the reported ownership effect concerns the mixed-label subset, not all 257 videos. Centroid and ridge candidates qualify on source. The protocol fixes eight donor shifts, four sentinels, 10,000 video-bootstrap draws, five strata, and source/confirmation roles. Subsequent crossed, wrong-owner, scene, and modern-model analyses use already consumed roles and remain post-confirmation evidence. Both learned candidates are reported, with simultaneous lower bounds across the two-candidate confirmation family; ridge is not presented as the only attempted candidate.

The dashes in Table II mean that ordinary frame utility was not a frozen promotion gate for these two historical prospective endpoints; they do not mean that utility was assumed to pass. OOPS was contracted as transition localization and used video-macro temporal AP inside the ownership contrast. Adding a pooled frame AUC/AP gate after its labels were opened would retroactively change the endpoint. For Pistachio, a later descriptive analysis reports confirmation-role raw frame AUC/AP of 0.7434/0.5826 for ridge, but those values are post-confirmation context and cannot be inserted into the original prospective decision. IITB was designed later and therefore predeclared both ordinary-utility and ownership gates.

B. One-Shot IITB Corridor Surveillance Study

IITB Corridor [22] provides a third independently frozen real-surveillance setting with 208 normal-training and 150 test videos. A metadata-only archive audit read no annotation payload bytes. Media-only extraction contained 358 AVI files and no NPY annotation files. Before opening annotations, we sealed three candidates: diagonal standardized distance from normal-training ResNet-18 embeddings, consecutive-embedding temporal novelty, and grayscale frame difference. The contract fixes eight donor shifts, four sentinels, 10,000 draws (seed 24082026), five strata, simultaneous three-candidate bounds, TIA/CNM floors, and separate AUC/AP promotion gates. Primary and replay prediction ZIPs are byte-identical.

The appearance candidate starts from 16 frames sampled uniformly from each training video. ImageNet-1K ResNet-18 embeddings define a diagonal standardized normal prototype; squared standardized distance is the score. Temporal novelty uses one minus cosine similarity between consecutive sampled embeddings, with fixed interpolation to native frame indices. Frame difference is mean absolute native-frame

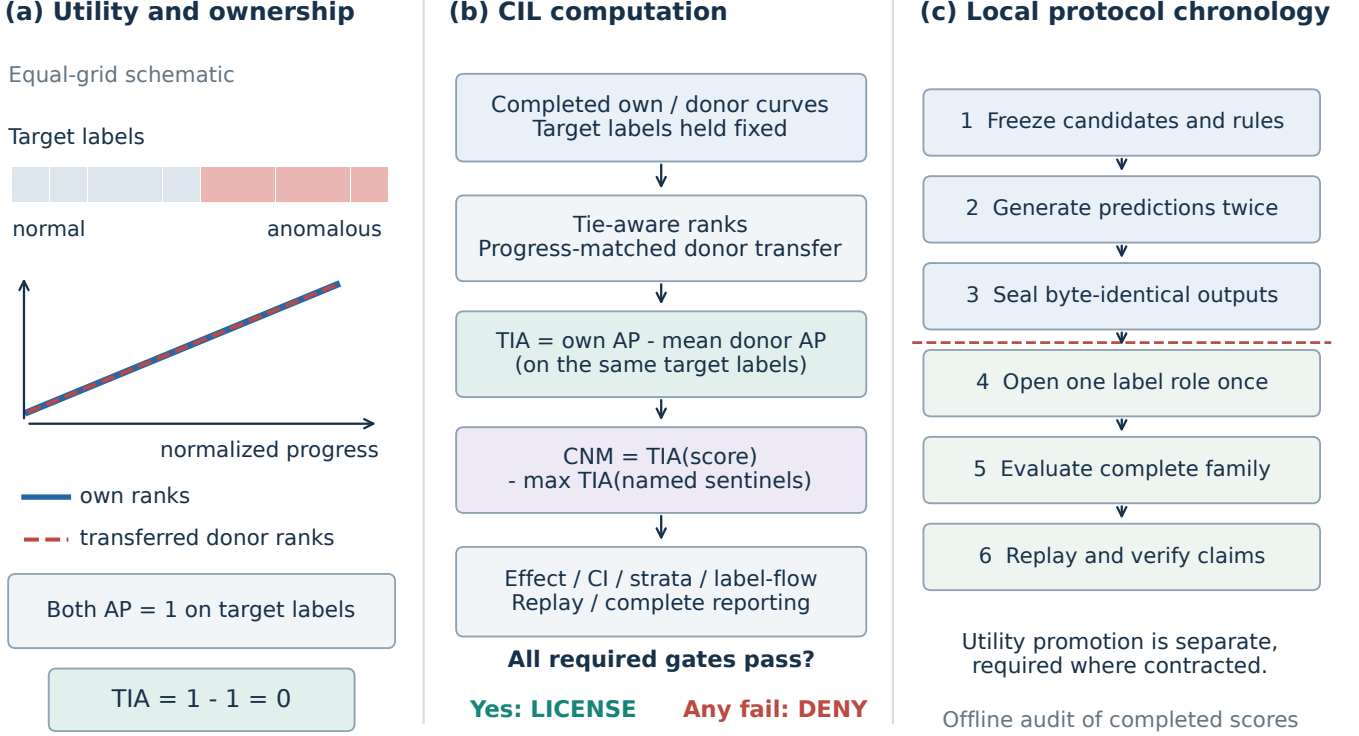


Fig. 1. CIL motivation, architecture, and fail-closed chronology. (a) Fixed target labels turn own versus progress-matched donor ranks into an ownership contrast that ordinary utility cannot provide. (b) CIL subtracts the strongest frozen sentinel advantage and applies effect-size, uncertainty, strata, label-flow, and replay gates; ordinary-utility promotion is separate and required where contracted. (c) The local protocol chronology has one label opening. Curves are mathematical schematics, not dataset examples.

grayscale difference divided by 255. All orientations are fixed as higher-is-more-anomalous. This heterogeneous family asks whether ownership survives in an appearance normal model, a representation-motion signal, and a parameter-free pixel signal; it was not selected to maximize an IITB leaderboard.

After one-shot label opening, all 150 annotation arrays were unanimously one element shorter than their decoded videos. No metric had been computed. An outcome-blind amendment therefore locked a common final alignment. The implemented evaluator drops score frame zero from all three candidates and requires exact length; it constructs the four analytic sentinels directly on that final grid. This differs from first constructing and then trimming a native-length sentinel, as detailed in the supplement. Frame zero is a constructed boundary for transition scores; the appearance-distance score is framewise, so applying this removal to it is an alignment convention. Drop-last, padding, interpolation, and outcome-selected or candidate-specific alignments were prohibited. This chronology is a disclosed protocol amendment, not represented as a rule frozen before label opening.

Of the 150 test annotations, 134 contain both normal and anomalous frames, 10 are all normal, and 6 are all anomalous. TIA is a within-target ranking contrast, so an all-constant label vector does not identify the required two-class temporal ordering; these 16 videos are excluded only as targets. They remain in the 150-video donor pool because donor construction is label independent. This yields 134 evaluable targets, 150 unique potential donors, and eight donors per target. Ordinary

pooled AUC/AP and prevalence use all 150 aligned videos, including the 16 constant-label videos. The 181,567-frame denominator therefore differs from the 134-target ownership denominator.

V. RESULTS

A. Prospective Confirmations

Table II reports the locked OOPS and Pistachio endpoints and the complete IITB family. OOPS latent motion attains CNM 0.0575 on 256 videos, and Pistachio ridge attains 0.1113 on 120 videos. IITB has 134 mixed-label evaluable targets; ordinary utility pools 181,567 aligned frames from all 150 videos. All three candidates have positive marginal and simultaneous TIA/CNM lower bounds and five of five nonnegative strata. Their AUC/AP pairs are 0.6818/0.7290, 0.7164/0.7523, and 0.6656/0.7102; all exceed the frozen AUC 0.60 and AP-over-prevalence 0.05 floors at prevalence 0.5964.

The sentinel family itself is uninformative or slightly adverse on IITB: center-time TIA is -0.00020 , duration-center is -0.00018 , linear progress is exactly zero, and sampling grid is 0.00088. The simultaneous one-sided TIA/CNM lower-bound pairs for normal distance, temporal novelty, and frame difference are 0.01562/0.01474, 0.04282/0.04194, and 0.02278/0.02190. Thus no conclusion depends only on an unadjusted marginal interval.

TABLE II

PROSPECTIVELY SEALED CONFIRMATION ENDPOINTS. IITB CORRIDOR REPORTS THE COMPLETE FROZEN CANDIDATE FAMILY. INTERVALS ARE 95% TARGET-VIDEO BOOTSTRAP CIs UNDER THE FROZEN DONOR SCHEDULE. n COUNTS MIXED-LABEL OWNERSHIP TARGETS: 120/257 FOR PISTACHIO AND 134/150 FOR IITB. IITB AUC/AP POOLS ALL 150 VIDEOS.

Study	n	TIA [95% CI]	CNM [95% CI]	AUC/AP	Decision
OOPS latent motion	256	.0583 [.0330,.0837]	.0575 [.0319,.0830]	–	licensed
Pistachio ridge	120	.1181 [.0899,.1465]	.1113 [.0809,.1413]	–	licensed
IITB normal distance	134	.0388 [.0158,.0620]	.0379 [.0149,.0611]	.682/.729	licensed + utility
IITB temporal novelty	134	.0660 [.0425,.0909]	.0651 [.0417,.0898]	.716/.752	licensed + utility
IITB frame difference	134	.0459 [.0252,.0678]	.0451 [.0244,.0666]	.666/.710	licensed + utility

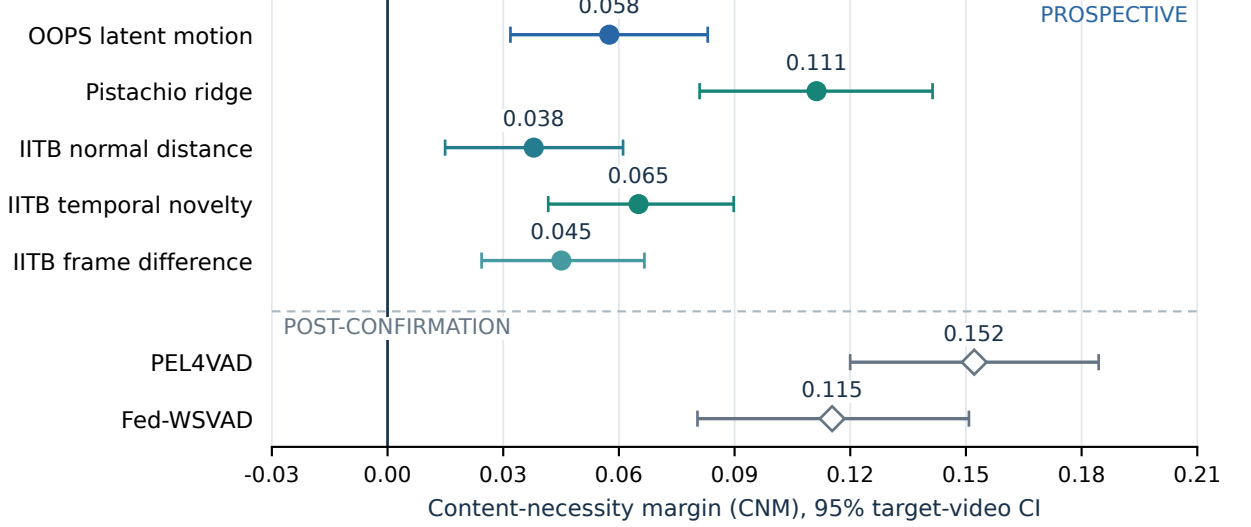


Fig. 2. Content-necessity margins across frozen evidence. OOPS, Pistachio, and all three IITB rows are prospective confirmations. PEL4VAD and Fed-WSVAD are post-confirmation pressure evidence only. Horizontal bars show 95% target-video bootstrap intervals under the frozen donor schedule. Compare signs and evidence roles; CNM is protocol-specific and is not a cross-dataset detector ranking.

B. Ownership, Side-Information, and Negative Controls

On the consumed Pistachio confirmation role, a crossed bootstrap over targets and donors retains ridge CNM 0.1113 with CI [0.0746, 0.1485]. Deterministic wrong-owner reassignment yields -0.0109 with CI $[-0.0369, 0.0148]$. All six eligible scenes retain nonnegative own-curve CNM. These analyses address donor reuse, arbitrary ownership, and single-scene explanations; they are not new confirmations.

We then applied a separately recorded two-way multinomial target/unique-donor bootstrap to the already consumed OOPS and IITB endpoints (5,000 draws; the sentinel maximum recomputed per draw). OOPS retains CNM 0.0575 with CI [0.0259, 0.0888]. IITB normal distance, temporal novelty, and frame difference retain CNM 0.0379 [0.0097, 0.0669], 0.0651 [0.0367, 0.0953], and 0.0451 [0.0205, 0.0705], respectively. The intervals widen under this resampling scheme, while every marginal lower bound remains positive. These are marginal robustness intervals, not simultaneous confidence bounds for the added family. Because these rules were added after the outcomes were known, they are explicitly post-confirmation robustness and do not replace the prospective video-only

intervals or add an independent confirmation.

The OOPS confirmation contains a nine-family panel; only frozen latent motion has a strictly positive confirmation interval. In the retrospective EgoOops falsifier, position/length side information has TIA 0.1095, exceeding V-JEPA2 ActionFormer TIA 0.0500. The candidate CNM is therefore -0.0595 and the license is denied. Four frozen parameter-free Pistachio dynamics candidates also fail source qualification; the closest CNM is 0.0136 with CI $[-0.0249, 0.0480]$. Positive ownership is not presented as a generic property of every score family.

C. Modern-Model and Retrospective Surveillance Evidence

PEL4VAD [23] has positive post-confirmation CNM but misses its frozen 0.75 AUC floor (0.7268 source, 0.7430 confirmation role); it remains a scientific negative. Fixed-final-round Fed-WSVAD [24] passes utility and CIL gates with confirmation-role CNM 0.1153, but its implementation uses one local epoch and is not claimed as an exact reproduction of an upstream ten-local-epoch or test-selected configuration.

A separate retrospective ShanghaiTech contract froze three existing score arrays before computing CIL, but labels and

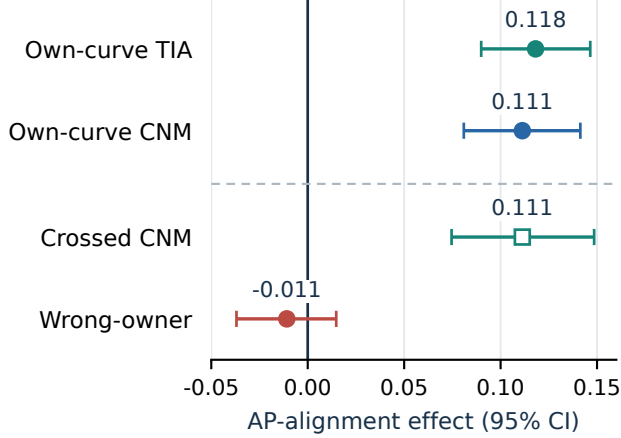


Fig. 3. Pistachio ownership control. The own-curve identity and sentinel-relative effects are positive, whereas deterministic wrong-owner reassignment yields an interval spanning zero; this does not prove a zero population effect. The first two rows repeat the prospective endpoint; crossed and wrong-owner rows are post-confirmation checks on that consumed role. Bars show 95% intervals.

ordinary metrics had already been used. Of 52 evaluable targets, ConvLSTM prediction has TIA/CNM 0.1378 and reconstruction 0.0999; both have positive simultaneous lower bounds but weak ordinary utility. Latent surprise is denied at CNM -0.0013 . The full family shows that target ownership and detector quality are non-equivalent; it cannot be called an untouched confirmation.

Table III makes two non-equivalences visible. First, a curve can be target-owned but have inadequate ordinary utility, as in PEL4VAD and the two positive Shanghai curves. Second, applying CIL to a contemporary score generator does not establish protocol-faithful reproduction of that generator. The score-family audit and detector-reproduction audit are separate claim objects.

D. Ordinary Utility and Ownership Answer Different Questions

Pooled frame AUC and AP depend on cross-video score ordering and frame weighting. Both are invariant to one shared strictly increasing score transform, while per-video transforms can change cross-video comparisons. AP also depends on class prevalence; ROC AUC is not intrinsically prevalence-dependent when the class-conditional score distributions are fixed. CIL computes a video-macro contrast after within-video ranking and donor transfer. High utility can therefore coexist with low ownership when a common progress shape transfers well, while positive ownership can coexist with weak utility when a poor detector produces target-specific variation. The IITB promotion rule requires both axes to avoid using ownership as a substitute for detection. The ShanghaiTech and PEL4VAD rows remain valuable precisely because they show the failure of that substitution.

This distinction also limits cross-study comparisons. CNM is an effect in video-level AP units under a particular donor

TABLE III
POST-CONFIRMATION OR RETROSPECTIVE PRESSURE EVIDENCE.

Method	AUC	AP	CNM	Classification
PEL4VAD	.743	.592	.152	licensed, low utility
Fed-WSVAD	.792	.619	.115	licensed, bounded fidelity
Shanghai latent	.448	.356	-.001	denied
Shanghai ConvLSTM	.550	.483	.138	CIL pass, low utility
Shanghai reconstruction	.467	.392	.100	CIL pass, low utility

and sentinel protocol; it is not a normalized performance score. We compare signs, uncertainty gates, and failure modes across studies, but do not rank detectors by CNM or pool heterogeneous protocols into a universal estimate.

VI. DISCUSSION

A. What the License Establishes

A CIL pass supports one bounded statement: under the specified donor transfer and named sentinels, the target's own completed score ranks its labels better than progress-matched foreign curves by more than the strongest sentinel identity advantage. It does not identify the pixels, objects, causal events, or semantics that generated the score. This distinction matters in multimedia systems because temporal scores are often normalized, interpolated, or pooled per video before evaluation.

The three prospective studies are complementary rather than exchangeable. OOPS is real-camera transition localization with a historical single sentinel; Pistachio provides synthetic controlled breadth and crossed donor analysis; IITB adds one-shot fixed-camera real surveillance, complete-family simultaneous inference, and ordinary utility gates. We do not pool their effects or tune one protocol using another's outcomes.

B. Alternative Explanations and Reviewer-Visible Tests

Conventional AUC/AP tables leave several rival explanations open. A centered anomaly schedule can inflate utility without content; the sentinel maximum tests the named schedules directly. A finite donor pool can understate uncertainty; crossed resampling adds donor variation. An arbitrary ownership convention could create a positive effect; deterministic reassignment tests wrong owners. One synthetic scene could dominate, so the six predeclared eligible Pistachio scenes are reported separately. Post-outcome selection could favor one candidate; complete-family reporting retains failed parameter-free, low-utility, and denied branches.

These controls do not exhaust explanations. They make the strongest plausible ones visible and falsifiable under frozen rules. A reviewer can therefore disagree with donor compatibility or sentinel coverage without having to infer which branches were attempted and suppressed.

C. Limitations

Donor compatibility defines the estimand and remains an assumption. CNM covers only named sentinels; camera identity, scene structure, proposal density, duration organization, and other metadata may contribute. IITB is one site and camera

regime with high anomaly prevalence, and its candidates are deliberately simple instruments rather than a survey of current VAD architectures. The alignment amendment is structurally unanimous and outcome-blind but occurred after label-file opening. Broad surveillance generalization requires another unused site, prospectively frozen modern detector families, and independent operational replication.

D. Reproducibility, Data Rights, and Responsible Use

Every headline value is bound to a frozen result artifact and SHA-256 manifest; the renderer reads those artifacts without recomputing decisions. Exact replay means byte-identical outputs in the recorded environment, not third-party replication. The review artifact excludes videos, labels, restricted per-video predictions, credentials, and upstream weights; public or research access is not treated as redistribution permission.

CIL is not a deployment-ready safety, surveillance, or intent-inference system. It should not be used alone for consequential decisions about people. It audits a score representation and does not establish semantic understanding, physical causality, exhaustive deconfounding, or improved detection.

The evidence chain records contracts, asset and label receipts, prediction manifests, evaluator source, result seals, claim checks, and figure-input hashes. The associated property and integration tests cover the core calculations, refusal modes, and crossed implementation. Any public package remains aggregate-only pending author rights review: hashes prove local identity but do not grant redistribution rights. Here “prospectively sealed” refers to the recorded local sequence of contracts, prediction exports, and label opening. The record does not provide public preregistration, independent timestamp custody, or third-party witnessing. Nor does separate target/donor resampling model every dependency: the same recording can occur in both roles, and recordings can share a camera or scene. The reported intervals are conditional evidence under these assumptions, not a distribution-free guarantee.

VII. CONCLUSION

CIL makes temporal score-curve ownership testable by comparing a target’s own ranks with progress-matched foreign curves on the same labels and subtracting a frozen sentinel maximum. Three prospectively sealed settings answer yes for their audited endpoints. The IITB confirmation additionally reports a complete three-candidate real-surveillance family that clears predeclared utility gates. Falsifiers, ownership reassignment, failed candidates, and low-utility passes show why the decision must remain fail-closed. CIL is an auditable multimedia evaluation tool, not evidence that a detector understands anomalous behavior.

ACKNOWLEDGMENT AND AI-ASSISTANCE DISCLOSURE

Generative AI tools assisted research framing and critique, literature support, code and visualization development, manuscript drafting and editing, and internal reproducibility checks. They were not treated as authors or as authority for label access, authorship, data release, or submission. The

human authors must verify every statement, figure, citation, and executable artifact; approve the final disclosure; and take full responsibility. Funding, acknowledgment, and conflict text, if any, must be added before submission.

REFERENCES

- [1] W. Sultani, C. Chen, and M. Shah, “Real-world anomaly detection in surveillance videos,” in *Proc. CVPR*, 2018.
- [2] G. Pang, C. Shen, L. Cao, and A. van den Hengel, “Deep learning for anomaly detection: A review,” *ACM Comput. Surv.*, vol. 54, no. 2, pp. 1–38, 2021.
- [3] N.-C. Ristea, N. Madan, R. T. Ionescu, K. Nasrollahi, F. S. Khan, T. B. Moeslund, and M. Shah, “Self-supervised predictive convolutional attentive block for anomaly detection,” in *Proc. CVPR*, pp. 13576–13586, 2022.
- [4] A. Acharya, H. Sapkota, Q. Yu, and X. Liu, “The road less seen: Segment exploration for weakly supervised video anomaly detection,” in *Proc. CVPR*, pp. 14147–14156, 2026.
- [5] N. Tatbul, T. J. Lee, S. Zdonik, M. Alam, and J. Gottschlich, “Precision and recall for time series,” in *Proc. NeurIPS*, 2018.
- [6] B. Ramachandra and M. J. Jones, “Street Scene: A new dataset and evaluation protocol for video anomaly detection,” in *Proc. WACV*, 2020.
- [7] J. Zhou, K. McGuinness, J. Antony, and N. E. O’Connor, “A fine grained quality assessment of video anomaly detection,” in *Proc. CBMI*, 2022.
- [8] N. Rashvand, S. Yao, A. D. Pazho, B. R. Ardabili, and H. Tabkhi, “From frames to events: Rethinking evaluation in human-centric video anomaly detection,” in *Proc. CVPR Workshops*, 2026.
- [9] Y. He *et al.*, “VALU: A benchmark for video anomaly temporal localization and understanding at multiple semantic levels,” in *Proc. ACL*, pp. 1252–1296, 2026, doi: 10.18653/v1/2026.acl-long.56.
- [10] M. Cho, T. Kim, M. Shim, D. Wee, and S. Lee, “Towards multi-domain learning for generalizable video anomaly detection,” in *Proc. NeurIPS*, 2024.
- [11] Q. Liu and J. Paparrizos, “The elephant in the room: Towards a reliable time-series anomaly detection benchmark,” in *Proc. NeurIPS Datasets and Benchmarks*, 2024.
- [12] M. S. Sarfraz, M.-Y. Chen, L. Layer, K. Peng, and M. Koulakis, “Position: Quo vadis, unsupervised time series anomaly detection?” in *Proc. ICML*, 2024.
- [13] T. Ji, N. Chakraborty, A. Schreiber, and K. Driggs-Campbell, “An expert ensemble for detecting anomalous scenes, interactions, and behaviors in autonomous driving,” *Int. J. Robot. Res.*, 2025.
- [14] A. Del Giorno, J. A. Bagnell, and M. Hebert, “A discriminative framework for anomaly detection in large videos,” in *Proc. ECCV*, 2016.
- [15] R. T. Ionescu, F. S. Khan, M. Georgescu, and L. Shao, “Unmasking the abnormal events in video,” in *Proc. ICCV*, 2017.
- [16] Y. Liu, C.-L. Li, and B. Póczos, “Classifier two sample test for video anomaly detections,” in *Proc. BMVC*, 2018.
- [17] P. Good, *Permutation, Parametric and Bootstrap Tests of Hypotheses*. New York, NY, USA: Springer, 2005.
- [18] B. Efron and R. Tibshirani, *An Introduction to the Bootstrap*. New York, NY, USA: Chapman and Hall, 1993.
- [19] A. B. Owen, “The pigeonhole bootstrap,” *Ann. Appl. Stat.*, vol. 1, no. 2, pp. 386–411, 2007, doi: 10.1214/07-AOAS122.
- [20] D. Epstein, B. Chen, and C. Vondrick, “Oops! Predicting unintentional action in video,” in *Proc. CVPR*, 2020.
- [21] J. Li, H. Cai, M. Dong, M. Pu, S. You, F. Wang, and T. Huang, “Pistachio: Towards synthetic, balanced, and long-form video anomaly benchmarks,” *arXiv:2511.19474*, 2025, accepted at ECCV 2026.
- [22] R. Rodrigues, N. Bhargava, R. Velmurugan, and S. Chaudhuri, “Multi-timescale trajectory prediction for abnormal human activity detection,” in *Proc. WACV*, 2020.
- [23] Y. Pu, X. Wu, L. Yang, and S. Wang, “Learning prompt-enhanced context features for weakly-supervised video anomaly detection,” *IEEE Trans. Image Process.*, 2024.
- [24] B. Wang, C. Huang, J. Wen, W. Wang, Y. Liu, and Y. Xu, “Federated weakly supervised video anomaly detection with multimodal prompt,” in *Proc. AAAI*, vol. 39, no. 20, pp. 21017–21025, 2025, doi: 10.1609/aaai.v39i20.35398.
- [25] R. Geirhos, J.-H. Jacobsen, C. Michaelis, R. Zemel, W. Brendel, M. Bethge, and F. A. Wichmann, “Shortcut learning in deep neural networks,” *Nature Mach. Intell.*, vol. 2, pp. 665–673, 2020.
- [26] A. D’Amour *et al.*, “Underspecification presents challenges for credibility in modern machine learning,” *J. Mach. Learn. Res.*, vol. 23, no. 226, pp. 1–61, 2022.

- [27] I. Song and J. Lee, "Frame-level evaluation in weakly supervised video anomaly detection mostly measures video-level ranking," *arXiv preprint arXiv:2608.21854*, 2026.
- [28] S. Abdulaziz and E. Bondarev, "Auditing frame-level AUC in weakly supervised video anomaly detection: Granularity, resolution, and scene bias," *arXiv preprint arXiv:2608.11985*, 2026.
- [29] M. Rashidi, "Benchmark AUC is not deployable reliability: A cross-dataset audit of off-the-shelf features for surveillance video anomaly detection," *arXiv preprint arXiv:2606.29506*, 2026.